Thejus P Aditya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.32
Infant Jesus School	Class XII (CBSE)	2019	95.60%
L'ecole Chempaka	Class X (ICSE)	2017	94.67%

EXPERIENCE

ASIC Design Engineer -- Netrasemi Pvt. Ltd.

Chips to Startup (C2S) Project -- MeitY

- **SoC RTL & Integration:** Developed and integrated RTL for a **RISC-V based SoC** targeting edge-AI applications, integrating **GPIO, PLIC, DMA, UART, I2S, SPI, I²C, Timer, and CPI peripherals**.
- **AMBA & Interconnect:** Designed and integrated **AMBA-based bus interfaces and SoC interconnects**, supporting communication between processor, memory, and peripheral subsystems.
- **Verification & Sign-off:** Performed **RTL linting, CDC analysis, and functional debugging** using **Synopsys SpyGlass**, supporting **tapeout-readiness** and design verification.
- **Version Control:** Managed RTL development and project workflows using **Git-based version control**.
- **FPGA Prototyping:** Performed **Xilinx Vivado-based FPGA prototyping and debugging** on the **Nexys Video** development platform.

SKILLS

Coding: Verilog, System Verilog, C++, Python

EDA Tools: Cadence Virtuoso, XILINX Vivado, LT-Spice, Cadence Xcelium, Cadence Simvision, Icarus Verilog

Domains: RTL Design, FPGA Design, ASIC Design, SoC Design and Integration, Soc Validation, Computer Architecture ,ML

RELEVANT COURSES

Digital Domain: CMOS Digital Design, Digital IC Design Lab,Microprocessor and Computer Architecture

Analog Domain: Analog IC Design, Analog IC Design Lab

Devices and Fabrication: Nanoelectronics Principles and Devices, Mesoscopic Device Electronics, Semiconductor Devices and Modelling

Memory: Memory Circuits and SystemDesign

PROJECTS

M-Tech Thesis: FPGA-based Hardware Accelerator for UAV Object Detection

Guide: Dr. Shishir Kumar

- Developing an **FPGA-based hardware accelerator for UAV object detection at the edge** using the lightweight **SkyNet CNN architecture**.
- Developing an **end-to-end FPGA implementation integrating a camera interface with the CNN accelerator** for real-time image acquisition and object detection.
- Validating the accelerator through **C/C++ simulation, RTL simulation, and hardware-oriented verification**.

CORDIC-Based Divider -- RTL Design & Simulation

- Designed and implemented a 16-stage **CORDIC-based hardware divider** using **vectoring and rotation modes**.
- Developed a **doubly pipelined architecture** to improve throughput and enable efficient hardware operation.
- Performed **RTL design, functional simulation, and verification** of the divider.

8-Stage Multi-Modulus Divider -- RTL Design & Verilog Implementation

- Designed and implemented an **8-stage multi-modulus divider** using cascaded **2/3 divider cells.**
- Extended the divider range to a **continuous programmable range of 2-511** using multiplexers.
- Implemented **programmable MI masking and frequency selection** with RTL simulation and verification.

CMOS-Based Serial Data Processing and 3T-DRAM Memory Array

- Designed a **serial data receiver and sequence detector** using 8-bit SIPO and start/stop detection for **4-bit payload.**
- Designed and integrated a **4×4 3T-DRAM memory array** with 2×4 decoder and **skewed CMOS sensing buffer.**
- Verified the design in **Cadence Virtuoso (GPDK90)** with **20 ns/bit reception and 1 ns memory operations**.

Real-Time Power Prediction using Machine Learning

- Implemented **Linear Regression, XGBoost, and Random Forest** models for real-time power prediction.
- Collected **INA sensor measurements and hardware performance counters** from Odroid and Raspberry Pi platforms.
- Developed and evaluated **ML-based power prediction models** for real-time system power monitoring.

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA): Embedded Systems

PUBLICATIONS & PRESENTATIONS

- Presented **"Integrated Testing Framework for DMA Peripherals"** at the **2024 IEEE SPICES International Conference**, IIIT Kottayam.

EXTRACURRICULAR

Exploring nature through **cycling, macro photography and outdoor adventures**.